

Compromised Air Traffic Control Workstation

Case Reference: INC-2026-0624-0001

Examiner: niel · 2026-06-24 22:29 UTC

3

CRITICAL

4TOTAL
FINDINGS**9**

INDICATORS

01.0 EXECUTIVE SUMMARY

Incident Overview

Concise narrative of the incident — what happened, when, impact, and required actions.

3

CRITICAL SEVERITY

YESCOMPROMISE
CONFIRMED**19 min**

TIME TO C2

Phishing

ATTACK VECTOR

KEY FINDINGS

On 2025-07-22, an air traffic control workstation (TSO-ATC-CT412, user 'award') was compromised within 3 minutes of user login via a phishing-delivered trojan (epxplorer.exe) masquerading as explorer.exe. The malware established command-and-control to an attacker-operated Telegram Bot API behind Cloudflare infrastructure 19 minutes after execution. At the time of compromise, ATC_RADAR.exe was actively processing aviation radar data.

Required Action: Immediate network isolation of TSO-ATC-CT412 · Escalate to aviation safety officer · Potential ICAO Annex 17 reporting obligation

Event Reconstruction

Chronological sequence of events reconstructed from memory forensics artifacts. Each event is traced to a specific tool and finding reference.

TIME (UTC)	DATE	EVENT	TYPE	SOURCE TOOL	FINDING REF
18:50:46	2025-07-22	System boot	SYSTEM	volatility3 windows.info	—
18:50:50	2025-07-22	User 'award' logs in — explorer.exe (PID 5116) starts	USER	volatility3 windows.pslist	—
18:51:58	2025-07-22	Microsoft Edge establishes HTTPS connection (browsing activity begins)	USER	volatility3 windows.netscan	—
18:53:11	2025-07-22	MALWARE EXECUTION: epplorer.exe (PID 9920) spawned by explorer.exe	MALWARE	volatility3 windows.pslist + MemProcFS	F-niel-001
19:12:33	2025-07-22	C2 ESTABLISHED: epplorer.exe → 104.21.1.247:443 (Telegram Bot API)	C2	volatility3 windows.netscan	F-niel-002
19:15:05	2025-07-22	ATC_RADAR.exe active — malware coexisting with aviation operations software	PROCESS	volatility3 windows.pslist + MemProcFS	F-niel-004
19:18:53	2025-07-22	Memory dump captured — evidence acquisition point	SYSTEM	volatility3 windows.info	—

Type legend: Malware Network/C2 Process System User

Detailed Investigation Findings

Each finding includes the observation, interpretation, impact assessment, MITRE ATT&CK mapping, tool traceability, and remediation guidance.

F-NIEL-001

CRITICALHIGH

Malware Execution: epexplorer.exe Masquerading as explorer.exe

Observation: Process epexplorer.exe (PID 9920) was spawned by explorer.exe (PID 5116) at 2025-07-22 18:53:11 UTC, just 2 minutes 21 seconds after user login. The binary was executed from C:\Users\award\Downloads\epexplorer.exe.

Impact: The malware achieved code execution on an air traffic control workstation within 3 minutes of user login. The filename explicitly targets the Windows shell process (explorer.exe → epexplorer.exe, substituting 'l' with '1') to evade casual observation in Task Manager. This is a targeted attack against an aviation-critical system.

MITRE ATT&CK	TOOL	EVIDENCE	CROSS-VALIDATION
T1036.005, T1204.002	volatility3 2.7.0 + MemProcFS 5.17.8	EVID-001 · raw/vol3_pslist.csv	MemProcFS proc.txt confirmed identical process tree

F-NIEL-002

CRITICALHIGH

Command and Control via Fraudulent Telegram Bot API

Observation: epexplorer.exe established an outbound HTTPS connection to 104.21.1.247:443 at 2025-07-22 19:12:33 UTC — 19 minutes after execution. The destination IP is a Cloudflare edge node fronting the attacker-controlled domain api-telegram-org.ctf.do. The malware uses Telegram Bot API (token: 8169267144:AAFwkHmXh71SMVcvFLantjTlwlRbT9HdJdU) as its C2 channel, abusing legitimate cloud infrastructure to blend with normal HTTPS traffic.

Impact: Active C2 channel established. Attacker has bidirectional communication with the compromised host. Telegram API provides file upload/download capabilities — data exfiltration or secondary payload delivery is highly probable. Cloudflare proxying makes IP-based blocking ineffective.

MITRE ATT&CK	TOOL	EVIDENCE	CROSS-VALIDATION
T1102.002, T1573	volatility3 2.7.0	EVID-001 · raw/vol3_netscan.csv	None — network capture unavailable for encrypted traffic analysis

Phishing Delivery Vector: Browser Download of Malicious Executable

Observation: epexplorer.exe was discovered at three virtual addresses in the memory filescan, including a partial browser download artifact (epexplorer.exe435568.crdownload). The .crdownload extension is specific to Chromium-based browsers (Microsoft Edge in this environment — confirmed by extensive Edge artifacts throughout the filescan). Per CTF intelligence, the phishing email originated from ktso.sec@mailforce.net.

Impact: User 'award' received a phishing email, clicked a link, and downloaded the malware via Microsoft Edge. The .crdownload artifact proves this was a browser-initiated download, not a drive-by or automated deployment. User awareness training gap identified. The phishing sender domain (mailforce.net) should be investigated for additional campaigns targeting the organization.

MITRE ATT&CK	TOOL	EVIDENCE	CROSS-VALIDATION
T1566.001, T1204.002	volatility3 2.7.0	EVID-001 · raw/ vol3_filesca.csv	None

Critical Asset Compromised: Air Traffic Control Workstation

Observation: The compromised host (TSO-ATC-CT412, user 'award') runs ATC_RADAR.exe — an air traffic control radar application with multiple active processes and network connections to 8.8.8.8 (Google DNS-over-HTTPS). The workstation name (ATC-CT412) and application context confirm this is not a standard enterprise endpoint — it is an aviation operations system. Multiple ATC_RADAR.exe instances ran concurrently with epexplorer.exe, meaning the malware coexisted with active ATC operations.

Impact: This is a worst-case compromise scenario. An attacker gained code execution on a safety-critical aviation system while it was actively processing radar data. Even if the malware's primary purpose was reconnaissance or C2 relay, the mere presence of unauthorized code on an ATC workstation constitutes a potential aviation safety incident. Regulatory reporting obligations may apply (ICAO Annex 17, national aviation authority).

MITRE ATT&CK	TOOL	EVIDENCE	CROSS-VALIDATION
T1003.001	volatility3 2.7.0 + MemProcFS 5.17.8	EVID-001 · raw/ vol3_pslist.csv	Dual-tool corroboration

Actionable IOCs

Blockable and searchable indicators extracted from findings. Use these to hunt for additional compromised hosts and block attacker infrastructure.

TYPE	INDICATOR	CONTEXT	SOURCE FINDING
SHA-256	3dc0d114859c0bde08d39155eaaa8f76392dd5121ca44ecb15652b3bf6049e35	Evidence archive	EVID-001
SHA-256	7b45e797acabaf592784bf7efceb979e3f036d5b9dd47f52280c17e38864c2dd	Memory dump	EVID-001
MD5	75ea94b54420c39dcd3d8ce574ba9d34	epxlorer.exe hash	F-niel-001
Filename	epxlorer.exe	Malware binary	F-niel-001
File Path	C:\Users\award\Downloads\epxlorer.exe	Execution path	F-niel-001
IP	104.21.1.247	C2 server	F-niel-002
Domain	api-telegram-org.ctf.do	C2 domain	F-niel-002
Email	ktso.sec@mailforce.net	Phishing sender	F-niel-003
IP	172.17.4.12	Compromised host	F-niel-004

Target System Identification

System metadata extracted from the memory dump. This is contextual information, not a finding.

PROPERTY	VALUE
Hostname	TSO-ATC-CT412
Username	award
Os	Windows 10 build 19041 (19041.1.amd64fre.vb_release.1912)
Architecture	x64
Processors	4
Host Ip	172.17.4.12
Dump Time	2025-07-22 19:18:53 UTC
Kernel Base	0xf80617e00000
Nt Product Type	NtProductWinNt
Source Tool	volatility3 2.7.0 windows.info.Info

Chain of Custody

All evidence items with cryptographic hashes and acquisition metadata.

EVIDENCE ID

EVID-001 – BelkaCTF_7_CASE250722_KTS0AERO.zip

SHA-256

3dc0d114859c0bde08d39155eaaa8f76392dd5121ca44ecb15652b3bf6049e35

SOURCE

https://dl.ctf.do/BelkaCTF_7_CASE250722_KTS0AERO.zip

Containment & Eradication

Prioritized remediation actions per finding. Execute in severity order (Critical → High → Medium).

F-NIEL-001

CRITICAL

Malware Execution: epexplorer.exe Masquerading as explorer.exe

Immediately isolate TSO-ATC-CT412 from all networks. Kill PID 9920. Quarantine C:\Users\award\Downloads\epexplorer.exe. Do NOT reboot before capturing full disk image — volatile evidence in memory will be lost.

F-NIEL-002

CRITICAL

Command and Control via Fraudulent Telegram Bot API

Block outbound HTTPS to api-telegram-org.ctf.do at network perimeter. Block Telegram Bot API endpoints if not business-justified. Deploy SSL inspection for traffic to unknown Cloudflare IPs. Check firewall logs for additional connections from 172.17.4.12 in the 18:50-19:18 UTC window.

F-NIEL-003

HIGH

Phishing Delivery Vector: Browser Download of Malicious Executable

Search mail server for emails from ktso.sec@mailforce.net — identify all recipients. Purge the phishing email from all inboxes. Block mailforce.net at email gateway. Initiate user awareness training for 'award' and ATC personnel. Review Edge browser security policies — consider disabling automatic execution of downloaded files.

F-NIEL-004

CRITICAL

Critical Asset Compromised: Air Traffic Control Workstation

Escalate to aviation safety officer and CISO immediately. This may require mandatory reporting to the national aviation authority. Ground the workstation — do not allow it to process live radar data until forensic imaging is complete and the system is rebuilt from trusted media. Engage the ATC software vendor (ATC_RADAR.exe) to verify software integrity. Review separation requirements between administrative workstations and operational ATC systems.

Limitations & Unresolved Questions

Areas not yet examined. Each gap represents uncertainty that should be addressed before final conclusions.

GAP 01 — DISK IMAGE NOT ANALYZED

Without filesystem forensics, persistence mechanisms and file-system timeline cannot be confirmed.

GAP 02 — NETWORK CAPTURE UNAVAILABLE

Encrypted C2 traffic cannot be analyzed for exfiltration volume.

GAP 03 — REGISTRY HIVES NOT EXAMINED

User activity artifacts and persistence locations remain unexamined.

GAP 04 — MALWARE NOT REVERSE-ENGINEERED

Full capability assessment of explorer.exe is pending.

GAP 05 — SUPPLEMENTARY EVIDENCE NOT ANALYZED

The 5.6 GB supplementary file was not analyzed.

GAP 06 — SINGLE HOST ANALYSIS ONLY

Compromise of additional systems cannot be ruled out.

Forensic Toolchain

Tools, versions, and runtimes used during this investigation. All tools validated by session canary before evidence analysis.

TOOL	VERSION	RUNTIME	VALIDATION STATUS
volatility3	2.7.0	docker	VALIDATED
memprocfs	5.17.8	host_fuse	VALIDATED
plaso	20240512	docker	VALIDATED
mft-tools	1.2.0.0	docker	VALIDATED
sift-vm	?	?	VALIDATED
session_canary	?	?	DEGRADED

Evidence Artifacts — Raw Tool Output

Unmodified tool output captured during analysis. Each artifact is traceable to the findings register and provides the raw data behind every conclusion in this report.

The following evidence screenshots were captured from raw tool output during analysis. Each image is an unmodified rendering of the tool's stdout/stderr. Original files are preserved in `case/raw/`.

ARTIFACT A01
01
231KB PNG · case/raw/screenshots/artifact-01.png

ARTIFACT A02
02
152KB PNG · case/raw/screenshots/artifact-02.png

ARTIFACT A03
03
94KB PNG · case/raw/screenshots/artifact-03.png

ARTIFACT A04
04
119KB PNG · case/raw/screenshots/artifact-04.png

ARTIFACT A05
05
261KB PNG · case/raw/screenshots/artifact-05.png

ARTIFACT A06
06
263KB PNG · case/raw/screenshots/artifact-06.png

ARTIFACT A07
07
264KB PNG · case/raw/screenshots/artifact-07.png

 View all: [case/raw/screenshots/](#) · Gallery: [case/raw/screenshots/index.html](#)

EXAMINER SIGNATURE

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This report is auto-generated from structured case data. All findings require human review before finalization.